

Examiner reports

Q1.

In part (a), the vast majority of students recognised the need to complete the square, with varying degrees of success. Most were able to remove a factor of 2, but less were able to

deal with the algebraic manipulation to achieve the required $\left(x - \frac{5}{4}\right)^2$. One third of all students went on to obtain the correct final result in the given form. Some students expanded and equated coefficients, but did not always put their result back in the required form.

In part (b), students did not always use inequalities as required. A variety of approaches were employed, including using the y coordinate of the vertex, forming a quadratic and using the discriminant and calculus.

Q4.

- (a) Provided candidates began by writing $b^2 - 4ac \geq 0$, and did not write the coefficient of x incorrectly by writing $b = k - 2$ instead of $-(k - 2)$, then full marks were often earned. Some wrote $b^2 = -(k - 2)^2$, however, and then conveniently “lost” the minus sign. There were not many sign errors, possibly because the required inequality was given in the question.
- (b) There were several completely correct solutions. Those who chose to factorise the equation to find the critical values were usually more successful than those using the quadratic equation formula, as $\sqrt{48^2 - 4 \times 7 \times 80}$ defeated many without the use of a calculator. Candidates are advised to draw a sketch or sign diagram in order to solve a quadratic inequality. Some gave their final answer as $k \leq 4$, $k \geq \frac{20}{7}$ or $\frac{20}{7} < k < 4$, both of which were penalised by one mark.

Q5.

- (a) The students who understood the concept gained three marks easily, either by using two equations or by squaring both sides. A few solved all four relevant equations obtaining repeated values of x . However, it was disappointing to see many wrong approaches, particularly those which gave intercepts with the axes or answers such as $|x| = |3|$. The sketch was there to help; so values such as -3 should have been spotted as being clearly incorrect.
- (b) Many students spoiled their answers by giving combined inequalities such as $1^3 x^3 \leq 3$. Other incorrect responses included the interval $1 \leq x \leq 3$ or the use of “ $x \leq 1$ and $x \geq 3$ ”. Again, the sketch was there to assist, and it was good to see some students marking the diagram to help them choose the correct part.

Q6.

In part (a), the simple inequality was not always obtained; many tried to work backwards from the answer, and several confused perimeter and area.

In part (b), once again many tried to work backwards from the printed answer and omitted the vital step of writing the area as $x(x + 4)$, which had to be less than 96.

In part (c), generally factorisation of the quadratic was done well; however, a few wrote $(x - 12)(x + 8)$ as their factorised form. Those who wrote down an incorrect answer without working usually only scored marks for the critical values. Candidates are strongly advised to draw a sketch or sign diagram.

It was very rare to see a correct answer to part (d), with the majority of candidates not considering the condition $x > 5.5$. Many simply wrote down the inequality $-12 < x < 8$ again. One common misconception was that the possible values of the width of the garden had to be integers. Some gave 6 and 7 as their answer, while many listed all the integers from -12 to $+8$.

Q7.

Very few candidates scored full marks for this question.

Part (a) was well done with only a handful failing to recognise a stretch and most giving the correct direction and scale factor; most who were wrong in one of these were wrong in both. It was essential to use the correct term 'translate' for the other, so 'shift' did not earn marks and 'transformation' was inadequate. A few made an error in the vector but most were correct.

Most candidates were unsuccessful in answering part (b) of the question as their curve was either below the x -axis or continued into the second quadrant. Many placed the curve through e instead of $e + 1$, and the curvature, for $x < e + 1$, was often wrong.

Correct answers in part (c)(i) were sometimes spoiled by poor algebra. It was worrying to see the misunderstanding of the log function, with $\ln(x - e)$ expanded as $\ln x - \ln e$ and incorrect subsequent work once a correct answer had been found: $e^{-1} + e = 0$ was a common such response.

In part (c)(ii), although the first mark for $x \geq 2e$ was often earned (though $x \leq 2e$ was common), the second mark was only gained by a very few candidates, as the part of the inequality concerning e was generally omitted completely.

Q8.

In part (a)(i), the idea of y being increasing when $\frac{dy}{dx} > 0$ seemed unfamiliar to many students, who seemed content to substitute a few numerical values for x into the

expression for $\frac{dy}{dx}$ rather than tackling the three-line proof. Consequently, very few scored full marks on this part.

Many students ignored the inequality signs throughout part (a)(ii). It was pleasing to see many students attempting to factorise the quadratic, often successfully, rather than relying on the formula. Perhaps more students would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. There were many

correct solutions, but students must realise that $x < 2$, $x > \frac{4}{3}$ is not the same as $\frac{4}{3} < x < 2$.

In part (b)(i), most students attempted to evaluate $\frac{dy}{dx}$ when $x = 2$. Not all were able to cope with the evaluation of $-6(2)^2$, which they needed to do in order to convince

examiners that $\frac{dy}{dx}$ was indeed equal to zero. Once more, some failed to write a concluding statement or referred to a stationary point rather than the fact that the tangent was parallel to the x -axis.

In part (b)(ii), it was pleasing to see some fully correct solutions. Most recognised that the equation of the tangent at P was $y = 3$, but it was disappointing to see a number of students write $y - 3 = 9(x - 2)$ followed by $y = x + 1$ as their tangent equation. In finding

the equation of the normal at Q , some, having correctly found that $\frac{dy}{dx} = 10$ when $x = 3$,

used the value 10 instead of $-\frac{1}{10}$ for the gradient of the normal. Arithmetic errors were very common by those who used $y = mx + c$ when finding the value of c ; typically $y =$

$\frac{x}{10} + \frac{7}{10}$ was seen instead of

$y = \frac{x}{10} - \frac{13}{10}$. Curiously, quite a few students who had previously obtained $y = 3$ as the equation of the tangent, and consequently the correct y -coordinate of the point R , then substituted the correct value of x into their incorrect equation for the normal and found a different, and now incorrect value for the y -coordinate of R .

Q9.

In part (a)(i) the negative coefficient of x^2 confused many candidates. The correct answer was rarely seen. Those who chose to work initially with the negative of the given expression, even successfully finding $(x + 5)^2 - 29$, often wrote their final answer as $29 - (x - 5)^2$. Candidates need to practise completing the square for expressions of this type even though it has not previously been tested very often.

In (a)(ii) too many had no idea how the line of symmetry related to the previous part. Many candidates just wrote down the maximum point or the equation of the parabola.

In part (b)(i) although the majority of candidates realised that they should equate the two equations, numerous algebraic errors appeared before the final line and even this final equation was sometimes incorrect.

In part (b)(ii) even some able students were unable to cope with squaring $2(2k + 5)$ in order to find an expression for the discriminant. Again, poor algebra and fudging abounded in order to obtain the printed inequality. To earn all the marks it was essential to have stated the condition $b^2 - 4ac > 0$ early in their working.

In (b)(iii) Only correct factors or quadratic formula earned the first mark; candidates at this level should all be capable of this, particularly with 29 being a prime number. Unfortunately, even with correct factors, some lost the minus sign on one of their critical values. Many failed to draw the appropriate sketch or sign diagram for determining the

correct solution, and a few lost the final mark by trying to combine the two correct inequalities thus producing an intransitive statement

such as $-\frac{29}{4} > k > -1$.

Q10.

Many candidates ignored the inequality signs throughout this question.

In part (a) the linear inequality defeated many whose algebra was weak; multiplying out brackets is a skill that all AS-level candidates should have mastered before starting a more advanced course, but sadly this often is not the case. It was quite alarming to see the low success rate in this part of the question, with most candidates removing the brackets and writing $8 - 6x > 5 - 4x + 8$ and consequently scoring no marks. Those candidates who obtained $-2x > -11$ were seldom able to obtain the correct final inequality, forgetting to change the $>$ sign.

In part (b) the slightly unusual way in which the quadratic inequality was presented did faze a few, but most candidates found the correct critical values. Perhaps more candidates would be successful in solving quadratic inequalities if they were to draw a sign diagram or sketch a graph. Admittedly, bright candidates are able to write down the correct solution with little or no working, but quite a few spoiled a correct solution by writing their final answer as $1.5 \leq x \leq -4$.

Q11.

In part (a), most candidates found at least one of the correct coordinates for the centre C , with the most common error being at least one sign error or writing the centre as $(4, -12)$.

However, the correct value for the radius was not so common, with $\sqrt{15}$ being frequently seen.

In part (b), most explanations involving a comparison of the y -coordinate of the centre and the radius of the circle earned at least one mark. In order to score the second mark, some of the better answers explained why the y -coordinate of the “highest” point of the circle was -1 , but most comments were insufficient. Those who proved that the circle did not intersect the x -axis needed to state that the y -coordinate of the centre was negative in order to score full marks.

Part (c)(i) was very poorly done. Many candidates tried to use their circle equation and essentially faked the given result instead of correctly using the distance formula. Once again, if candidates are asked to “show that ...” then the full equation needs to be seen in the final line of the proof and an essential part of the working was to see a statement such as $PC^2 = 3^2 + (k + 6)^2$, leading to the printed answer.

In part (c)(ii), the word “hence” indicated that the expression for PC^2 in part (c)(i) needed to be used. Candidates needed to realise that the point P lies outside the circle when $PC > r$ and that using this result immediately leads to the given inequality. The inequality sign here caused confusion with many looking for a discriminant.

In part (c)(iii), many candidates scored only the two marks available for finding the critical values -2 and -10 . No doubt some would have benefited from practising the solution of

inequalities of this type by drawing a suitable sketch, or by familiarising themselves with the technique of using a sign diagram as indicated in previous mark schemes.

Q12.

Part (a) was well answered by the majority of candidates. The main errors were not crossing the y -axis into the second quadrant or failing to give the points of contact with the axes.

In part (b), most candidates obtained the required two values. In most cases where candidates lost marks, it was for only giving one value, usually 2. Four values of +2, -2, +6, and -6 was not uncommon.

Part (c) was not as well answered with many candidates trying to write down a single inequality for an answer.

Q13.

Part (a)(i) Candidates did not seem well drilled in completing the square when the coefficient of x^2 is not equal to 1. It was very rare to see a correct answer here although a few did realise that $p = 5$. Clearly further practice is required at this type of question.

Part (a)(ii) Only the more able candidates were able to reason sufficiently well using the result from part (a)(i) as well as providing a concluding statement. No credit was given for using the discriminant to show that the equation had no real roots since the wording of the question excluded this approach.

Part (b)(i) This kind of question has been set several times before and the usual errors were seen. Candidates should state the condition for real roots ($b^2 - 4ac \geq 0$) and find an expression in terms of k for the discriminant using the correct inequality throughout. The inequality is then reversed when multiplying by a negative number. Again, many candidates would benefit from practising this technique, using brackets where appropriate to avoid algebraic errors.

Part (b)(ii) The factorisation of the quadratic was usually correct, but several candidates

wrote down one of the critical values as $\frac{1}{7}$. Most found critical values and either stopped or immediately tried to write down a solution without any working. Candidates are strongly advised to use a sign diagram or a sketch graph showing their critical values when solving a quadratic inequality.

Q14.

In part (a) it was quite alarming to see the number of candidates who were unable to factorise this quadratic.

Most candidates scored only a single mark in part (b) for attempting to find the critical values. Many would benefit from practising the solution of inequalities of this type by drawing a suitable sketch or by familiarising themselves with the technique of using a sign diagram as indicated in previous mark schemes.

Q15.

In part (a), many candidates applied at least one correct formula, but it was not

uncommon to see $S_n = \frac{n}{2}(a + l)$ quoted and used as though l , the last term, was 1, or l not replaced by $a + (n - 1)d$. A significant number of average candidates attempted a verification or just used ' $38 = a + 24 \times 1.5$ ' to get ' $a = 2$ ' and then applied the same general

equation to get the circular argument ' $38 = 2 + 24d$ so $d = \frac{38 - 2}{24} = 1.5$ '.

Some better candidates correctly obtained the two equations ' $38 = a + 24d$ ' and ' $1250 = 20(2a + 39d)$ ' but then just wrote down ' $d = 1.5, a = 2$ '. As ' $d = 1.5$ ' is printed in the question, candidates were required to show sufficient working for the solution of these simultaneous equations before the final two marks could be awarded. The most common error in part (b) was candidates linking S_n to 100 rather than using $u_n < 100$. However, it was pleasing to see many of those candidates who could not answer part (a) using the printed value for d to answer part (b).

Q16.

Most candidates scored the mark for the correct printed equation in part (a), but some omitted " $= 0$ " and others made algebraic slips when taking terms from one side of their equation to the other.

In part (b)(i) only the more able candidates were able to obtain the printed inequality using correct algebraic steps. Many began by stating that the discriminant was less than 0, clearly being influenced by the answer. Not all assigned the correct terms to a , b and c in the expression $b^2 - 4ac$ and others made sign errors when removing brackets.

The factorisation was usually correct in part (b)(ii), but many wrote down one of the critical

values as $\frac{1}{3}$. Most found critical values and either stopped or immediately tried to write down a solution without any working. Candidates are strongly advised to use a sign diagram or a sketch showing their critical values when solving a quadratic inequality.

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Q17.

In part (a), some weaker candidates did not realise how to derive the given equation, and others made algebraic slips when proving the printed result, or failed to write " $= 0$ ".

In part (b), the condition for two distinct points of intersection required candidates to use the condition that $b^2 - 4ac > 0$ at any early stage of their argument. Those who simply wrote " > 0 " on their final line of working, without any previous reference to the discriminant being positive, failed to convince the examiners that they deserved full marks.

In part (c), quite a number were unable to factorise the quadratic correctly and many resorted to using the quadratic equation formula to find the critical values. Where this was done correctly but left in surd form, it was given due credit except for the final mark. Very able candidates can write down the answer to the inequality once they have factorised the quadratic but far too many guessed at answers and an approach using a sign diagram or sketch is recommended. Candidates also need to realise that the final form of the answer

cannot be written as $\frac{14}{9} < k < -2$.

Q18.

The most common error by far in part (a) was the omission of brackets in $(4k)^2$. Usually $16k^2$ appeared after $4k^2$ had been seen in the line above. Candidates must be far more careful when proving a given answer in order to convince the examiner. It is essential to address the inequality by stating the condition for real roots early in the proof, rather than just inserting it into the final line.

In part (b), factorisation of the quadratic proved too difficult for many, with the most common error being wrong signs in the brackets or the roots. Some used the quadratic formula, but the calculation of $(-9)^2 - 4 \times 4(-9)$ defeated many.

Of those who calculated the critical values correctly, many assumed that they had answered the question and stopped. It was good to see more candidates showing their method of solution of the inequality by drawing a sketch or sign diagram. It was a pity that many of those who obtained the correct answer then wrote their final answer as $3 \leq k \leq$

$$-\frac{3}{4}.$$

Q19.

Part (a) The condition for real roots was not widely known and the form of the printed answer caused many to write the condition as $b^2 - 4ac \leq 0$.

Part (b) It was disappointing to see many unable to factorise the quadratic correctly. Far too many guessed at answers and an approach using a sign diagram or sketch is recommended. Candidates also need to realise that the final form of the answer cannot be written as " $k \geq -5$ or $k \leq 8$ "

Q20.

Parts (a) and (b) were usually well done, but in part (b), although many completely correct answers were seen, there were also many diagrams where part (a) had been translated 4 units along the x or y axis and W-shape diagrams were also popular. Dealing with the modulus in part (c) proved to be challenging, although there were many fully correct

answers. Candidates appeared to be more successful with finding $x = \frac{4}{3}$ than $x = -4$. A

surprising number of candidates followed $3x = 4$ by $x = \frac{3}{4}$.

Those candidates who were successful in part (c) were usually successful in part (d) but correct values with incorrect inequality signs was a common error.

Q21.

Part (a) Only the more able candidates were able to complete this proof correctly. Many began by stating that the discriminant was less than or equal to zero, no doubt being

influenced by the printed answer.

Part (b)(i) The factorisation was usually correct.

Part (b)(ii) Most candidates found the critical values, but many then either stopped or wrote down a solution to the inequality without any working. Many candidates wrongly

thought the solution was $k \leq \frac{1}{2}, k \leq 2$. Candidates are advised to draw an appropriate sketch or sign diagram so they can deduce the correct interval for the solution.

Q22.

- (a) Not everyone recognised that the height of the rectangle was the value of y when $x = -1$ or $x = 4$. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.
- (b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.
- (ii) Almost everyone recognised that they should firstly evaluate the integral from -1 to 2 , but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it correctly.
- (c) (i) Differentiation was done well on the whole.
- (ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.
- (iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.
- $\frac{dy}{dx}$
 Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.
- (d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$ as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as -2 , $\sqrt{2}$, $1+\sqrt{2}$. Using the formula or completing the square sometimes led to $1 \pm \sqrt{1}$, which many candidates failed to simplify.

Q23.

- (a) It was disappointing to see a number of candidates unable to deal with the minus sign outside the second bracket. Another very common error was to simplify the expanded brackets correctly to $-11k^2 - 14k + 25$ and then to say that this was equal to $11k^2 + 14k - 25$, thus losing the accuracy mark.
- (b) (i) The sign in the printed answer caused many to write the condition for real roots as $b^2 - 4ac \leq 0$. There was a lot of 'fudging' in this part of the question and it was a rare event to see a candidate score full marks.
- (ii) The most successful candidates identified the critical values and then used a sign diagram or a sketch to identify the correct solution to the inequality.

Q24.

Part (a) It was quite disappointing to see the type of errors being made in this question. In trying to simplify the right hand side, $3 - 5(x + 6)$, many thought it was equal to $-2(x + 6)$; others wrote it as $3 - 5x + 30$ or $3 - 5x + 6$. Even after obtaining $8x > -24$, it was not unusual to see the final answer as $x < -3$.

Part (b) The majority of candidates were able to factorise the quadratic or use the formula to find the two critical points. It was expected that some method such as a sketch graph or sign diagram would have been used in order to obtain the final solution. Many wrongly asserted that $x < 3$, $x < -2$ and this scored no method marks for the inequality. Clearly some candidates were able to write down the correct solution after simply factorising the quadratic and scored full marks.